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Nonintrusive in-situ modeling for unobserved virtual models in digital twin-enabled building HVAC systems: A one-year comparison of data-driven and physics-based approaches in a living laboratory

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ABSTRACT

Virtual modeling of building systems presents significant challenges due to the prolonged operation under dynamic conditions. This study proposes a novel, nonintrusive in-situ modeling method to overcome the practical measurement difficulties while providing reliable long-term operational data. Previous research has achieved high-accuracy virtual models; however, their applicability was often limited to specific time periods or singular operating conditions. Building on these efforts, this study establishes an integrated framework that incorporates multiple virtual models developed through a sequence of prediction, benchmarking, and correction processes. To optimize model performance, data-driven and physics-based modeling approaches were employed for comparison. Field studies were conducted on a real heating, ventilation, and air conditioning system (HVAC), and long-term modeling performance was validated over a one-year period. The physics-based corrected model demonstrated superior performance, with the annual root mean squared error (RMSE) value significantly reduced from 0.592 m^3/h to 0.034 m^3/h, and the mean absolute percentage error (MAPE) decreased by 26.7 %. In summary, physics-based nonintrusive in-situ models offer excellent long-term applicability and facilitate optimal control and holistic monitoring of building system operations.

Nomenclatures

Abbreviations			
APE	Absolute percentage error	P	water-source heat pump power
BAS	Building management and automation system	$P(C\|Y)$	posterior distribution
BIM	Building information modeling	$P(Y\|C)$	likelihood function
DT	Digital twin	$P(Y)$	normalizing constant
HVAC	Heating, ventilation, and air conditioning	f	mathematical function
MAPE	Mean absolute percentage error	u	unknown parameter

(*continued on next page*)

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Abbreviations			
MCMC	Markov Chain Monte Carlo	x	input variable (*X: set of input variables)*
MLP	Multi-layer perceptron	x_c	correction parameter
RMSE	Root mean squared error	y	output variable (*Y: set of output variables)*
VBM	Virtual building model	γ	frequency ratio
		$\pi(C)$	prior distribution
Symbols		σ	standard deviation
c_w	specific heat capacity of water		
f_{pu}	pump frequency	Subscripts	
ΔT	temperature difference	*Dd*	data-driven
C	correction function	*Pp*	physics-based
$D(C)$	distance function	*b*	benchmark
DP	differential pressure	*cond*	condenser
E	thermal energy	*cor*	correction
G	mass flow rates	*eq*	equipment
H	pump head	*eva*	evaporator
i	value index	*obj*	objective function
ini	initial prediction	*r*	return water
m	sensor measurement	*s*	supply water
n	number of values	*sys*	system

1. Introduction

The global warming issues caused by the overproduction of greenhouse gases from power generation have become increasingly serious. With the rise in national living standards, the application of HVAC (Heating, Ventilation, and Air Conditioning) systems in the building sector has rapidly spread [1]. Due to the extensive use of operations, HVAC systems consume approximately 50 % of the total building energy to maintain the required indoor environment [2]. Additionally, unreasonable control sequences [3], inefficient set-points [4,5], and degradation of aging equipment [6,7] have led to various malfunctions [8,9], further contributing to increased energy consumption and the deterioration of indoor environment quality [10,11].

Building digitalization technology has been introduced to improve building performance and enhance operational efficiency. Building digitalization can be achieved through the concepts of Building Information Modeling (BIM), cyber-physical systems, and digital twins (DTs) [12,13]. These concepts leverage advanced sensing technologies, data-driven analytics, and machine learning to enable intelligent operational applications, thereby actively improving the overall performance of building systems [14]. In recent years, the concept of BIM and digital twins has been widely adopted and applied in the building sector. Petri et al. [15] proposed a digital twin-based adaptable architecture for resilient infrastructure, focusing on energy efficiency, sustainability, and advanced monitoring through wireless sensors in industrial and construction applications. Jiang et al. [16] explored the use of digital twins and semantic web technologies for intelligent control of building fire protection systems in fire accidents, integrating BIM and IoT data for enhanced virtual-real interaction. Spudys et al. [17] addressed deficiencies in operational building assessment practices by proposing the use of digital twin technology, BIM files, and smart sensors to connect physical and virtual data, enabling real-time energy performance assessment and advancing digital construction practices with a focus on energy efficiency indicators. Targeting building HVAC systems, Xie et al. [18] presented a digital twin-based approach using symbolic AI to detect HVAC faults by identifying key sensory data, labeling sensors with machine-readable fault tags, and automating real-time fault annotation and data filtering for dynamic asset management. Hosamo et al. [19] proposed the Digital Twin of the HVAC system method, integrating BIM, Simulink, and multi-objective genetic algorithm to optimize HVAC systems by processing IoT data, validating models, and identifying optimal energy and thermal comfort solutions, with results streamed back to BIM for implementation. Digital twins aim to accurately reflect physical objects in a digital form, span their life cycle, and use real-time data to analyze behaviors and operations [20]. The original digital twin model comprises three main elements: (1) the physical entity, (2) the virtual model, and (3) their real-time interactions for continuous data and information flow [21]. To create a comprehensive digital twin environment, virtual space should integrate various types of models, such as geometry, processes, physics, behaviors, and rules for building operations. By employing assembly, integration, and fusion techniques, the virtual model can describe and depict physical entities from multi-dimensional, multi-domain, and multi-scale perspectives. Such comprehensive virtual models are essential for digital twin applications and services, including virtual sensing [22], simulation [23], diagnosis [24], prediction [25], and optimization [26].

Focusing on the unique inherent and operational characteristics of building, Yoon [27] introduced the concept of Virtual Building Models (VBM), defined as mathematical models that simulate the physical behavior of operational buildings. Unlike other models within BIM and DT frameworks, which primarily focus on geometry and rules, VBMs emphasize mathematical, physical, and dynamic behavior models. These VBMs are composed of sub-virtual models (observed, unobserved, and designed), ranging from the initial designed model to fully detailed in-situ operational representations. With the objective of providing deeper insights into the behavior and operational status of real buildings, surpassing the limitations of textual, geometrical, and static information, VBMs must rely on in-situ sensor measurements during the modeling process. Therefore, sufficient sensors, datasets, and building domain knowledge are crucial to establishing a virtual model that accurately reflects real building operations. However, in real-world building systems, practical issues such as sensor absence, malfunctions, low redundancy, and sparse sensor deployment often limit the sensing environment [28]. As a result, a new modeling approach is essential to meet the conditions necessary for digital twin-enabled intelligent operations.

The concept of in-situ modeling methodologies [29] has been developed to establish a mathematical modeling process for building

behavioral models during real-time operations. In-situ modeling involves constructing mathematical behavioral models directly within the target building operations, aiming to produce more accurate models than those developed outside the target building environment (e.g., in different buildings, modeling environments, or time periods) [29]. This methodology accounts for the long operational life cycle of buildings, their large scale, on-site dynamic behaviors, and the challenges of replicating a laboratory environment for mathematical modeling. The in-situ modeling approach addresses uncertainties in physics-based models during their operational phase by applying mathematical correction techniques. Additionally, it incorporates sensor data to facilitate data-driven modeling, which is crucial for accurately representing real building operations. In this context, in situ modeling encompasses two main approaches: (1) intrusive in-situ modeling and (2) nonintrusive in-situ modeling. Intrusive in-situ modeling involves constructing virtual models, particularly for target models that are unobserved by sensors during building operations, by mathematically incorporating additional data obtained through intrusive efforts, such as measurements collected using specialized instruments operated by personnel within the building. Compared to the nonintrusive in-situ modeling approach, which relies solely on operational data collected by building management and automation systems (BAS), Intrusive modeling is relatively straightforward and accessible for non-professionals to understand and apply. However, conducting intrusive measurements during real operations is often challenging due to practical installation constraints and high external costs. Additionally, temporary sensors may introduce measurement errors, thereby increasing overall modeling uncertainties. Consequently, the key research focus is the development and validation of unobserved virtual models through non-intrusive approaches, thereby eliminating the reliance on additional intrusive measurements.

In the domain of building in-situ modeling, Wang et al. [30] employed an intrusive method to develop a principal components analysis model for the target sensor using early-stage accurate data. This approach enabled the detection and diagnosis of faults in temperature and pressure sensors. Xu et al. [31] diagnosed faults in temperature sensors and flow rate meters with varying amplitudes, using intrusive data to construct the sensor model. This method demonstrated improved performance in fault detection rate, diagnosis rate, and estimation accuracy. Yu et al. [32] applied an intrusive in-situ modeling technique for calibrating supply air temperature sensors, achieving a measurement uncertainty of 0.7 °C. Kaneko [33] demonstrated that nonintrusive thermal load monitoring could differentiate sub-loads generated by various heat sources (e.g., occupants, equipment, solar radiation) from the total thermal load, thereby allowing indirect assessment of a room's thermal properties without additional sensors. Koo et al. [34] proposed a nonintrusive in-situ modeling approach based on model fusion, which facilitated the calibration of the virtual return water temperature sensor, reducing systematic and random errors by 44.0 % and 32.9 %, respectively. Choi et al. [35] introduced a nonintrusive in-situ modeling method for virtual heating return water temperature sensors based on physical relationships and valve opening rates, resulting in a reduction of the root mean squared error (RMSE) by 0.26 °C, indicates the effectiveness of nonintrusive calibration methods. These recent studies have been increasingly focused on developing nonintrusive in-situ approaches due to their ability for automatic or autonomous modeling during building operations. To advance the nonintrusive in-situ modeling methods in building operations, previous studies [36,37] have developed virtual models for nonintrusive in-situ calibration using data-driven approaches.

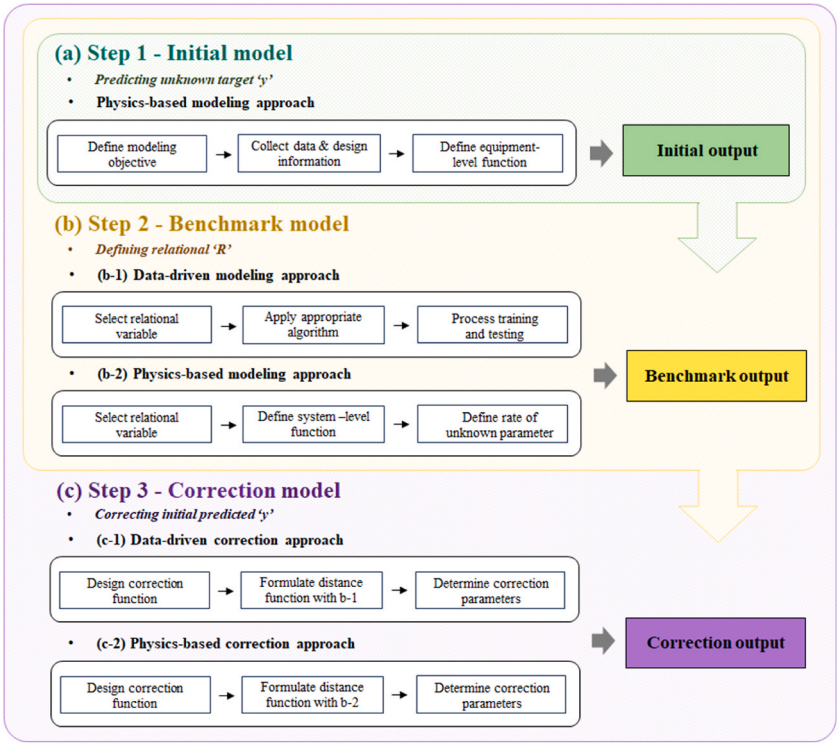


Fig. 1. Overall framework of nonintrusive in-situ modeling processes.

While data-driven corrected models have achieved reasonable accuracy, their performance remains highly sensitive to the hyper-parameters, and input features used during the modeling process. In this context, physics-based modeling presents significant potential for offering more robust and reliable performance, effectively addressing these limitations. Existing studies have demonstrated virtual modeling accuracy over limited operation periods (e.g., a few weeks, months, or a single season), however, as these models are functioned as the virtual equivalents of physical sensors without corresponding hardware, long-term reliability is crucial for the deployment of virtual models in real building operations. Furthermore, the developed modeling process must undergo rigorous evaluation to support the implementation of virtual models within digital twin environments.

Accordingly, this study aims to propose a nonintrusive in-situ modeling framework for unobserved virtual models in digital twin-enabled building HVAC systems, thereby bridging the gap between virtual modeling and digital twin environments. To enhance existing methodologies, a physics-based benchmark modeling approach is newly adopted into the nonintrusive modeling process to address the identified limitations in current research. this study aims to propose a nonintrusive in-situ modeling framework for. Case studies were conducted in a living laboratory for building HVAC systems, where data-driven and physics-based virtual models for chilled water flow rates were developed, refined, and validated over a one-year period. During the modeling process, BIM data played a pivotal role in establishing the prediction model as a robust foundation and providing reliable information for the development of benchmark models. The findings of this study have the potential to facilitate broader applications of virtual models, extending their use beyond traditional monitoring to encompass roles such as control feedback sensors or input sensors for intelligent applications in digital twin-enabled building operations. The rest of this paper is organized as follows: Section 2 outlines the overall framework and the three development steps for each modeling approach. Section 3 details the field application used for the comparative study. Section 4 presents the application results and discusses the model's effectiveness and potential usability for real-world applications. Finally, Section 5 concludes the paper.

2. Methodology

2.1. Overall framework

As shown in Fig. 1, the modeling process using the proposed nonintrusive in-situ virtual modeling method consists of three key steps: 1) initial model development, 2) benchmark model development, and 3) correction model development. In the absence of physical observations, the target variable is initially predicted using a white-box modeling approach based on physics and design information. A relational variable R, is introduced for benchmark and nonintrusive correction modeling processes. Benchmark models are developed to define R, and are constructed using data-driven or physics-based approaches, with the initial model serving as a mandatory input. The resulting benchmarks are compared to corresponding measurements of R, enabling the formulation of distance functions that indirectly measure the accuracy of the initial model. Finally, correction models are built based on these formulated distance functions within the nonintrusive calibration method, facilitating the determination of correction parameters. Notably, target measurements obtained from real sensors are not used throughout the modeling process. Instead, measurements from building management and automation system (BAS) (e.g., physical sensors, control signals), design information (e.g., equipment specifications, test reports), and system operation knowledge (i.e., physics) are essential references and inputs. Detailed explanations of each step are provided in the following subsections.

2.2. Development of initial prediction model

As shown in Fig. 1(a), the initial model is constructed in the first step to predict the unknown target y. In the absence of physical measurements, the target variable is defined using a white-box modeling approach that leverages BIM data, including building design information and equipment specifications, as input parameters. Standard value calculation sheets, along with mechanical, electrical, and plumbing plans, serve as critical references for the initial modeling process. The BIM data is extracted and derived based on the target variable and integrated into the corresponding operational physics equations. Rather than emphasizing prediction accuracy, the primary objective of the initial modeling is to capture the operational patterns of the target variable, thereby improving the performance of subsequent correction models. In this context, the equipment-level function (f_{eq}), which provides a direct mathematical representation of the equipment's performance and characteristics, is typically utilized as the governing equation, as shown in Eq. (1). It is important to note that achieving precise and accurate estimations at this stage is challenging due to the independent consideration of equipment and omission of various uncertainties present in system operations.

$$y_{ini} = f_{eq}(X_{eq}) \tag{1}$$

here, y_{ini} is the initial model output, f_{eq} is the equipment-level function and X_{eq} is the set of variables used in the equipment-level function.

2.3. Development of benchmark model

In the second step, data-driven and physics-based benchmark models are developed, as shown in Fig. 1(b). A relational variable R is introduced and employed as the modeling target. The objective of benchmark modeling is to define R in terms of initial model output, thus enabling the formulation of the indirect distance function required in Step 3. Among the existing variables in the target system, R is expected to exhibit the strongest correlations with the target variable. Due to the differing development concepts of data-driven and physics-based models, R is selected separately based on either intended physical relationships or data similarity analyses. The

performance of the developed benchmark model is crucial to the effectiveness of the virtual modeling and correction processes. Modeling accuracy is expected to improve as the corrected initial model output aligns more closely with the ground truth.

The data-driven (black-box) benchmark model is established using regression techniques, such as neural network-based machine learning algorithms, as shown in Fig. 1(b-1). The model is developed through training and testing processes to identify underlying patterns between inputs and outputs. Input variables are selected based on data similarities and prior experience [34], while the output is the measurement of the relational variable. The training results depend on the quality and quantity of the input datasets, the designed features and complexity of the model. Inaccurate or insufficient data can lead to modeling errors, and overfitting problems may occur due to repeated training. To mitigate these issues, the testing process is conducted to evaluate the trained model using additional datasets. The model outputs are compared against the provided data to identify significant discrepancies or poor generalization, thus ensure modeling accuracy [40].

The novel physics-based benchmark model is developed based on comprehensive building physics, as depicted in Figs. 1(b-2). A system-level function, such as energy and mass balance equations, that describes the physical relationships among multiple system variables, is employed as the governing equation. However, uncertainties inevitably arise in the system function due to theoretical and practical assumptions. To enhance the reliability of the physics-based benchmark model, an unknown parameter (u) is designed to represent overall operational uncertainties and is incorporated into the formulation as part of feature extraction, as shown in Eq. (2). This unknown parameter is initially defined based on research related to similar buildings or operating systems and is corrected simultaneously with the initial model in Step 3. Rather than directly correcting the initial model output, the use of u serves as an auxiliary approach to improve the performance of the physics-based virtual models during the correction processes.

$$y_b^{Pp} = f_{sys}\left(Y_{ini}, X_{sys}, u\right) \tag{2}$$

where, y_b^{Pp} is the output of physics-based benchmark model, f_{sys} is the system level function, X_{sys} is the set of system variables used as inputs to the system-level function and u is the unknown parameter.

2.4. Development of nonintrusive indirect correction model

In the final step, data-driven and physics-based nonintrusive correction models are constructed to indirectly correct the initial model output, as shown in Fig. 1(c). Conventional correction models for physical sensors are typically formulated using a distance function based on benchmark residuals, which account for the differences between benchmark values and sensor measurements. Building on this method, the distance function for correcting virtual models is obtained by coupling the model outputs with the corresponding physical observations, as shown in Eq. (3) [27,38].

A correction function C is designed with correction parameters X_c as shown in Eq. (4). The objective is to determine the most appropriate correction parameters that correct initial model outputs to better align with the target ground truth. However, the formulation of distance function in Eq. (3) is challenging due to the absence of target measurements as correction benchmarks. To overcome this difficulty, the nonintrusive correction method, which utilizes existing datasets to indirectly correct the initial model is therefore proposed. The objective distance function ($D(C)_{obj}$) is formulated using the benchmark model outputs from Step 2 and physical measurements of the relational variable, as shown in Eq. (5). Data-driven and physics-based correction models are developed based on the corresponding benchmark models, as shown in Fig. 1(c-1) and Fig. 1(c-2). $D(C)_{obj}$ is designed to be minimized when the most appropriate correction parameters are applied [41]. Finally, the initial model output is corrected by adding the correction value as shown in Eq. (6).

$$D(C) = (Y_b - Y_m)^2 \tag{3}$$

$$C = g(X, X_c) \tag{4}$$

$$D(C)_{obj} = (Y_b - Y_R)^2 \tag{5}$$

$$y_{ini}^{cor} = y_{ini} + C \tag{6}$$

3. Real application

3.1. Target HVAC system and sensing environment

The proposed nonintrusive in-situ modeling method was applied as a virtual sensor to obtain the mass flow rate on the evaporator side (G_{eva}) in a real HVAC operation system as depicted in Fig. 2. The configuration of the living laboratory includes an office room, a meeting room, and a student office room. The target water-source heat pump system is centralized, two cooling water pumps (with backups) and two chilled water pumps (with backups) are positioned on opposite sides. Detailed system information is provided in Table 1.

The air-conditioning system operates in cooling, heating, or all-fresh-air modes to achieve and maintain the user-set indoor temperature in the experimental area. Throughout the year, the system operates in cooling mode during summer (June to September), heating mode during winter (November to March) and all-fresh-air mode during transition seasons (April to May). In cooling mode, water circulates through the evaporator to absorb heat from the Air Handling Unit and dissipates through the condenser to the cooling

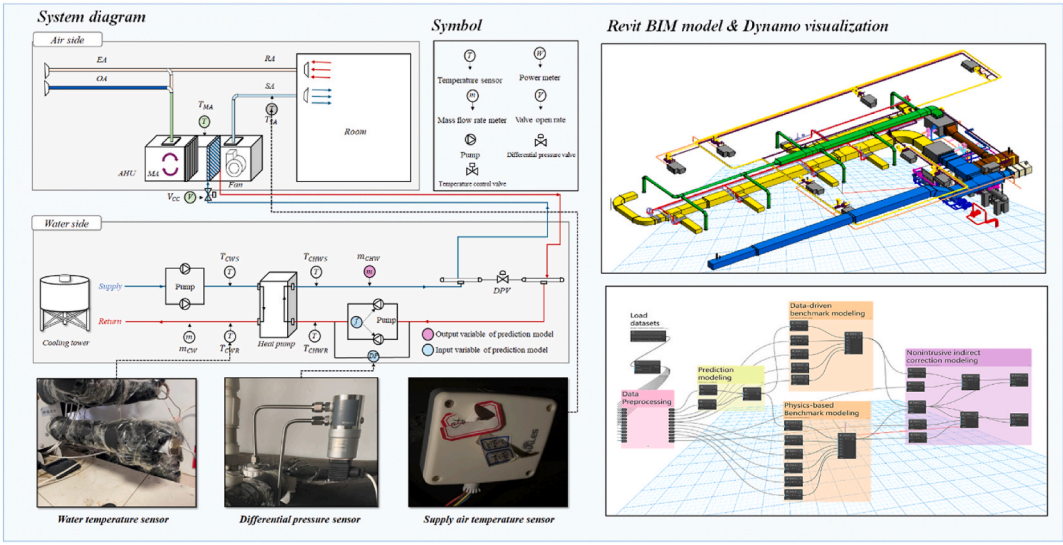


Fig. 2. Target building HVAC system.

Table 1
Detailed information about the target system.

Basic information	Component	Description
Building information	Building function	Experiment building
	Building structure	Concrete shear wall and steel structure
	Architectural orientation	North-South direction
	Heating mode	Central air conditioning system
	Cooling mode	Central air conditioning system
	Shape coefficient of building	0.60
	Window-wall ratio	0.19
VAV system information	Water source heat pump	Heat capacity 29.5 kW, input power 4.8 kW
	Water pump	Rated flow 2.7 m^3/h, delivery lift 12 mH_2O
	Air handling unit	Nominal air flow rate 5000 m^3/h, input power 1.92 kW
	VAV box 1/2/5/6	Measurement range 198–660 m^3/h
	VAV box 3/4	Measurement range 336–1120 m^3/h
	Air side temperature sensor	−40 - 70 °C, accuracy 0.6 °C
	Water side temperature sensor	Accuracy 0.04 °C
	Differential pressure sensor	0–400 kPa, accuracy 2 kPa

tower. The process is reversed in heating mode by adjusting the direction of the four-way reversing valve, while the cooling tower is replaced by a heat exchanger connected to the public heating system. In all-fresh-air mode, the main system is switched off, the chilled water pump operates independently to supply fresh air and improve indoor air quality.

The related system variables include supply and return temperatures on the evaporator side (T_{s-eva}, T_{r-eva}), pump frequency (F_{pu}), differential pressure value (DP_{eva}) and thermal energy on the condenser side (E). Sensor measurements were recorded using BAS at 1-min intervals over a one-year period (June 2022 to May 2023). Note that October 2022 was excluded due to irregular operations caused by extended public holidays and a lack of occupants.

3.2. Design of initial prediction model

The initial G_{eva} model was developed by applying the principles of pump affinity law, as shown in Eq. (7) and Fig. 3. The formulation was derived through pump characteristic curve fitting techniques, with pump frequency and differential pressure as input variables. The relationship between head and mass flow rate was expressed using polynomial regression, and the coefficients were calculated based on the reference values of head and pressure provided in the pump specifications, as detailed in Table 2. This regression function was then extended to account for operational pump frequency, using the standardized frequency of 50 Hz as a reference, as shown in Eqs. (8), (9) and (10). Consequently, the polynomial equation was generalized using the ratio of the measured frequency to the designed frequency, as illustrated in Eqs. (11) and (12). However, several uncertainties remained due to the following factors:

1. Lack of considerations of system operations.
2. Practical gaps between the pump specifications and real-world operations.
3. Unknown uncertainties beyond the formulated model.

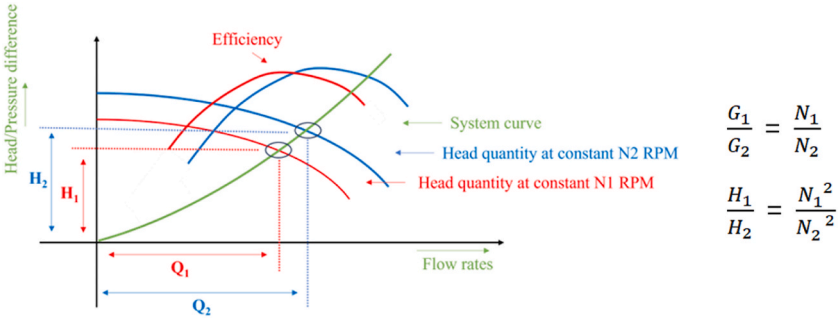


Fig. 3. Principles of the pump affinity-law.

Table 2
Pump performance specifications.

Variables	Measuring points									
Mass flow rate (G) (m^3/h)	0	1.2	1.8	2.4	3	3.6	4.2	4.8	6	7.2
Head (H) (m)	20.5	17.8	16.2	14.4	12.3	9.8	6.9	–	–	–

Here, N_1, N_2 are the constant pump speeds under different operation conditions, G_1 is flow rate with pump speed N_1, H_1 is the head/pressure difference with pump speed N_1, G_2 is flow rate with pump speed N_2 and H_2 is the head/pressure difference with pump speed N_2.

$$y_{ini} = f_{eq}\left(h(F_{pu}, DP_{eva})\right) \tag{7}$$

$$H_{pu,50} = \alpha_1 G_{pu,50}^2 + \alpha_2 G_{pu,50}^1 + \alpha_3 \tag{8}$$

$$y_{G_{eva}} = G_{pu,50} \times \left(\frac{F_{pu}}{f_{50}}\right) \tag{9}$$

$$H_{pu} = H_{pu,50} \times \left(\frac{F_{pu}}{f_{50}}\right)^2 \tag{10}$$

$$\gamma = \left(\frac{F_{pu}}{f_{50}}\right) \tag{11}$$

$$H_{pu} = \left\{\alpha_1 \left(y_{G_{eva}} \times \frac{1}{\gamma}\right)^2 + \alpha_2 \left(y_{G_{eva}} \times \frac{1}{\gamma}\right) + \alpha_3\right\} \times \gamma^2 \tag{12}$$

3.3. Benchmark model design

The data-driven benchmark model was developed using a multi-layer neural network, with the initial model output and related system variables as inputs. Pump power (P) was selected as the relational variable based on data similarity analyses, as suggested by Choi et al. [43]. A multi-layer perceptron (MLP) was trained using operational data of June 2022. To enhance modeling efficiency, data preprocessing techniques such as simple moving average, z-score outlier removal, and min-max normalization were applied. The MLP was structured with four hidden layers and activation functions. The Rectified Linear Unit (ReLU) was applied to the first three hidden layers, while a linear function was used for the output layer. Random shuffling and early stopping rules were implemented to mitigate overfitting issues [36]. Detailed information about the MLP data preprocessing, structure and hyperparameter settings is provided in Table 3.

The physics-based benchmark model was constructed using the balance equation between the evaporator and condenser sides of the target water-source heat pump system, as shown in Eq. (13). Thermal energy on the condenser side (E) was selected as the relational variable based on the intended physical relationships, and G_{eva} was substituted by the initial model output, as indicated in Eq. (14). The unknown parameter u was used to represent overall energy losses during operation and was defined based on research into uncertainties of building operational energy balances. Li et al. [39] reported a thermal imbalance rate of approximately 11.6 % when analyzing sensor faults during long-term operation. Guo et al. [42] found that the confidence interval for the balance equation in a new heat pump system was 90 %. Based on these findings, the unknown parameter u was initially estimated as a mode value of 10 %. Consequently, the physics-based benchmark model was formulated by applying u as a multiplication factor and expressed as shown in Eq. (15).

Table 3
MLP inputs and settings of the data-driven benchmark model.

Category	Setting/Approach
Data preprocessing	
Noise reduction	Simple moving average
Outlier removal	Z-score outlier removal
Scale consistency	Min-max normalization
Structure and hyperparameter	
Input layer	$y_{ini}, G_{cond}, T_{r-eva}, T_{s-eva}, T_{r-cond}$
Hidden layers	4 layers
Activation function	ReLU (hidden layers), Linear (output layer)
Batch size/Epochs	32/100
Loss function	Mean squared error
Optimizer	Adam
Training/ Validation Ratio	85 %/ 15 %

$$y_b^{Pp} = c_w G_{eva} \Delta T_{eva} + P \tag{13}$$

$$y_b^{Pp} = c_w y_{ini} \Delta T_{eva} + P \tag{14}$$

$$y_b^{Pp} = (1-u)(c_w y_{ini} \Delta T_{eva} + P) \tag{15}$$

3.4. Design of nonintrusive indirect correction model

As shown in Eqs. (16) and (17), the distance functions for the data-driven and physics-based correction models were formulated using benchmark model outputs from Step 2 and the measurements of the corresponding relational variables. The correction functions were designed as half-degree rational functions with two correction parameters x_{c1} and x_{c2}, as expressed in Eqs. (18) and (19). Additionally, an extra correction parameter, x_u, was determined simultaneously with x_{c1} and x_{c2} to adjust the unknown parameter u in the physics-based correction process. Since u was not included in the correction function, as discussed in Section 2.3, the corrected models were consistently obtained by adding the correction function to the initial prediction, as described in Eq. (20).

The correction parameters were estimated using Bayesian inference, as detailed in Eqs. (21)–(23). The posterior probability (Eq. (21)) was derived from the prior probability, $\pi(C)$, the likelihood function (Eq. (23)), and the normalizing constant, as shown in Eq. (22). Priors were assumed to follow a Gaussian distribution with a mean of 0 and a standard deviation of 0.5, reflecting the uncertainty in the initial model's residuals. The likelihood function was defined as a conditional probability distribution with a zero-mean Gaussian distribution. The normalization constant, $P(Y)$, was computed using the Markov Chain Monte Carlo (MCMC) method, with the Metropolis-Hastings algorithm employed for sampling [44,45]. The parameters x_{c1}, x_{c2} and x_u were obtained by minimizing the distance function and maximizing the posterior probability $P(C|Y)$ during the inference process. A summary of the time schedules for both virtual modeling processes is presented in Table 4. The correction models were developed at the beginning of the modeling period, and the corrected model outputs were validated and analyzed throughout the remainder.

$$D(C)^{Dd} = \left(Y_b^{Dd} - Y_P\right)^2 \tag{16}$$

$$D(C)^{Pp} = \left(Y_b^{Pp} - Y_E\right)^2 \tag{17}$$

$$C = f^{cor}(Y_{ini}, X_c), X_c = [x_{c1}, x_{c2}] \tag{18}$$

$$f^{cor}(Y_{ini}, X_c) = x_{c1} Y_{ini}^{1/2} + x_{c2} \tag{19}$$

$$Y_{ini}^{cor} = Y_{ini} + x_{c1} Y_{ini}^{1/2} + x_{c2} \tag{20}$$

$$P(C|Y) = \frac{P(Y|C) \times \pi(C)}{P(Y)} \tag{21}$$

$$P(Y) = \int P(Y|C)\pi(C)dx \tag{22}$$

$$P(Y|C) = \frac{1}{\sigma\sqrt{2\pi}} exp\left[-\frac{1}{2\sigma^2} D(C)\right] \tag{23}$$

here, Y_b^{Dd} is the data-driven benchmark model output, Y_P is the measurement of power, Y_b^{Pp} is the physics-based benchmark model output, Y_E is the thermal energy on condenser side, $P(C|Y)$ is the posterior distribution, $P(Y)$ is the normalizing constant, $\pi(C)$ is the

Table 4
Virtual modeling scenarios and schedules.

Scenarios	Periods	
	Physics-based	Data-driven
Prediction modeling period	06/01/2022–05/30/2023	
Benchmark training period	/	06/01/2022-06/30/2022
Benchmark testing period	/	07/01/2022-07/04/2022
Correc**tion modeling period**	06/16/2022–06/20/2022	07/05/2022-07/09/2022
Validation period	06/21/2022-05/30/2023	07/10/2022-05/30/2023

prior distribution, $P(Y|C)$ is the likelihood function, and σ is the standard deviation.

4. Results and discussions

The long-term performance of the established virtual models was evaluated over the entire validation period, from June 2022 to May 2023. Modeling accuracies were quantified by addressing the uncertainty relative to the target ground truth data. Root mean squared error (RMSE), absolute percentage error (APE), and mean absolute percentage error (MAPE) values were calculated as performance indicators to facilitate comparison and analysis, as shown in Eqs. (24)–(26). It is important to note that the ground truth dataset for G_{eva} was segregated and used exclusively for evaluation and validation purposes.

$$RMSE = \sqrt{\frac{\sum_{i}^{n} (y_i - \hat{y}_i)^2}{n}} \tag{24}$$

$$APE = \frac{|y_i - \hat{y}_i|}{Y_i} \tag{25}$$

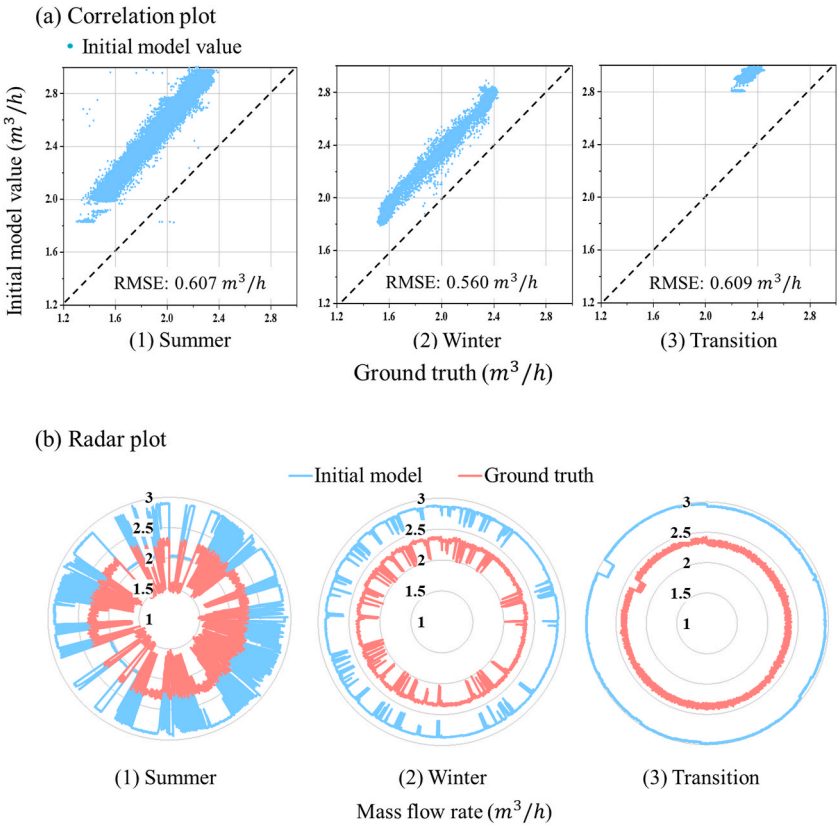


Fig. 4. Seasonal performances of the initial model.

$$MAPE = \frac{\sum_{i=1}^{n} \frac{|y_i - \widehat{y_i}|}{Y_i}}{n} \tag{26}$$

where, y_i is the index calculation referenced value, $\widehat{y}_i$ is the value obtained from model, i is the value index, and n is the total number of values.

4.1. Performance of initial prediction model

The comparison between the initial model outputs and the ground truth, along with RMSE calculations, is presented in Fig. 4. G_{eva} was generally overestimated, with notably high seasonal RMSE values throughout the year. The primary and common reason for these significant uncertainties is the omission of heat loss and lack of consideration for actual building conditions. In real operations, heat loss through water flow is unavoidable, and system configurations are often adjusted to meet specific construction and usage requirements. The seasonal RMSE values were calculated as 0.607 m^3/h (summer), 0.560 m^3/h (winter), and 0.609 m^3/h (transition), indicating minimal differences across the seasons. However, substantial differences were observed in the distribution patterns of the data. As shown in Fig. 4(a), the model exhibited poor performance during the summer season, characterized by the widest data distribution range. In contrast, the distribution range narrowed in the winter season. Notably, during the transition season, a unique pattern emerged, where the majority of data points were concentrated in a specific region.

The observed performance differences can be attributed to unstable operating conditions. The valve opening rate fluctuated frequently between 30 % and 100 % in response to the comparison of the supply-air temperature with its setpoint. These variations directly resulted in significant changes in G_{eva}, thereby increasing modeling complexity and reducing estimation accuracy. As depicted in Fig. 4(b), the system exhibited the highest valve adjustment rates in summer, leading to the most unstable operational status. Conversely, during the winter season, the system operated more stably with reduced variations and uncertainties. During the transition season, when neither heating nor cooling was required, the system was turned off, and the pump operated independently. Under these conditions, the valve remained fully open without fluctuations, resulting in consistent system operation throughout the season.

The inconsistent seasonal performance of the initial model underscores the challenges of achieving long-term reliability using a single model. These findings highlight the necessity of implementing lateral, non-intrusive correction processes to improve model performance and adaptability under varying operating conditions.

4.2. Performances of data-driven benchmark and corrected models

The training and testing results of the data-driven benchmark model were compared, as shown in Fig. 5. As shown in Fig. 5(a), significant errors were observed at the beginning of the training period, particularly with negative values occurring between 500 and 1000 episodes. These errors can be attributed to the nature of the data-driven modeling approach, which lacks consideration of physical principles. Consequently, the model may produce unrealistic outputs that deviate from real-world behavior, especially under extreme conditions. Better performance was observed during the testing period, with RMSE values decreased from 0.43 W to 0.37 W, indicating the importance of the training process on data-driven modeling accuracy. However, despite these overall improvements,

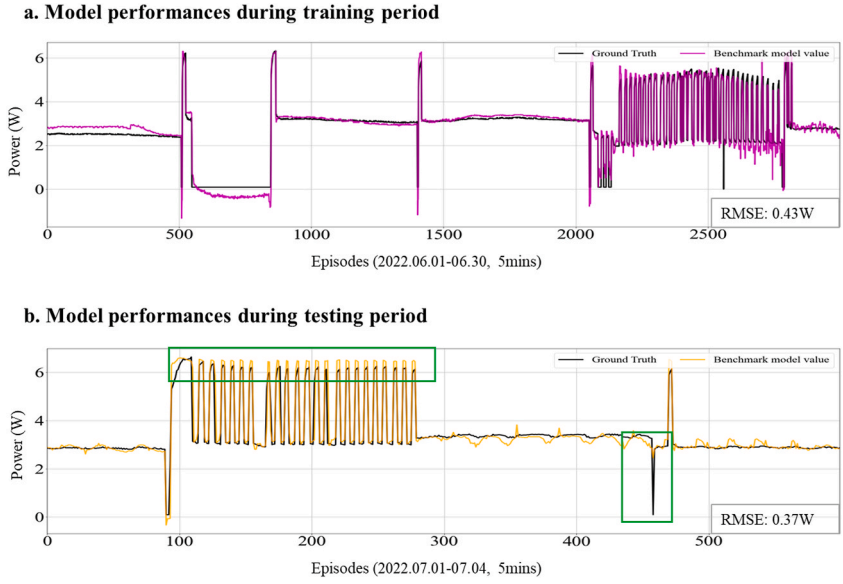


Fig. 5. Comparisons of data-driven benchmark model accuracy between training and testing period.

significant errors were still present at the upper and lower extremes of the data range, as depicted in Fig. 5(b). This issue arises from an inherent limitation of data-driven approaches, where the model's output is highly dependent on the data distribution in the training dataset. When the model is trained with limited extreme data, predictions tend to converge towards the majority range (i.e., between 1.5 and 2.5 W), leading to overestimations at lower power values and underestimations at higher power values.

The long-term validation results of the data-driven corrected model are presented in Fig. 6. The RMSE and MAPE values were significantly reduced from 0.59 m^3/h to less than 0.11 m^3/h and from 30 % to within 4.2 %, respectively, demonstrating substantial improvement over the entire validation period. However, notable seasonal inconsistencies were observed. After training and testing process, the data-driven model performed best during the summer season, achieving the lowest MAPE of 2.15 %, while performance declined during the winter and transition seasons, with MAPE peaking at 4.17 %. These seasonal declines highlight the limitation of data-driven correction models, which rely on training patterns rather than physical interactions, making them unable to self-adjust to operational variations. Consequently, additional training and correction processes are necessary for winter and transition seasons to maintain long-term modeling accuracy.

4.3. Performances of physics-based benchmark and corrected models

The improving performance of the physics-based benchmark model is depicted in Fig. 7. As shown in Figs. 7(a-1) and (a-2), the MAPE value decreased from 6.18 % to 4.66 % after applying the manually defined unknown parameter u of 10 %. The resulting data distribution pattern was observed to align more closely with the ground truth. Additionally, the APE values during the correction period were compared, revealing an overall reduction, as shown in Figs. 7(b-1) and (b-2). These results highlight the usefulness of incorporating the unknown parameter and emphasize the importance of accurately determining the appropriate u to optimize benchmarking accuracy.

Through the Bayesian inference process, as illustrated in Fig. 8, the correction parameters were determined when the posterior probability was maximized. The resulting values of -0.147, -0.256 for x_{c1} and x_{c2}, respectively, and 0.516 for x_u. Fig. 9 demonstrates that the accuracy of the calculated mass flow rate from the initial prediction model was consistently and significantly improved. Unlike the data-driven corrected results shown in Fig. 8, the physics-based model not only maintained high modeling accuracy across all seasons but also demonstrated enhanced performance during winter and transition periods. Specifically, the modeling accuracy was measured using RMSE and MAPE values, which were 0.041 m^3/h and 1.79 %, respectively, during the summer season. These values decreased in the winter season and reached a minimum of 0.026 m^3/h and 0.89 % during the transition season. The operational stability of the pump emerged as the key factor influencing uncertainty in the physics-based modeling. The best performance was observed during the transition season, when G_{eva} operated predominantly at high flow rates. The sustained high modeling accuracy was achieved through the integrated structure of multiple white-box sub-models, with uncertainties effectively mitigated by constraining the correction model using various operation physics.

By applying x_u to the manually defined u, the physics-based benchmark model was corrected, as shown in Figs. 7(a-3) and (b-3). The model outputs closely aligned with the ground truth, with 89.7 % of the results achieving low uncertainties of less than 5 %, based on APE values. The MAPE value was minimized to 2.37 %, resulting in an overall accuracy improvement of 61.7 % to the original

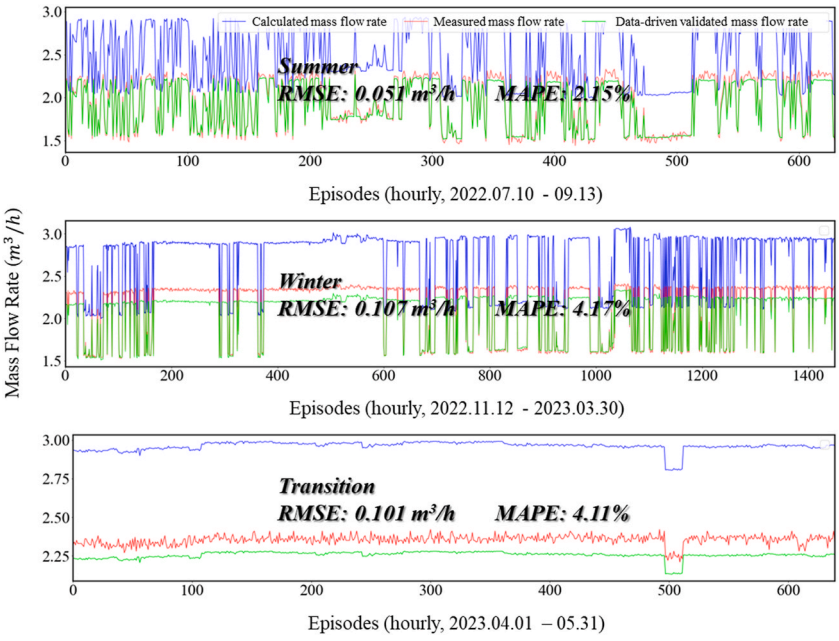


Fig. 6. Long-term validation results of the data-driven corrected target model.

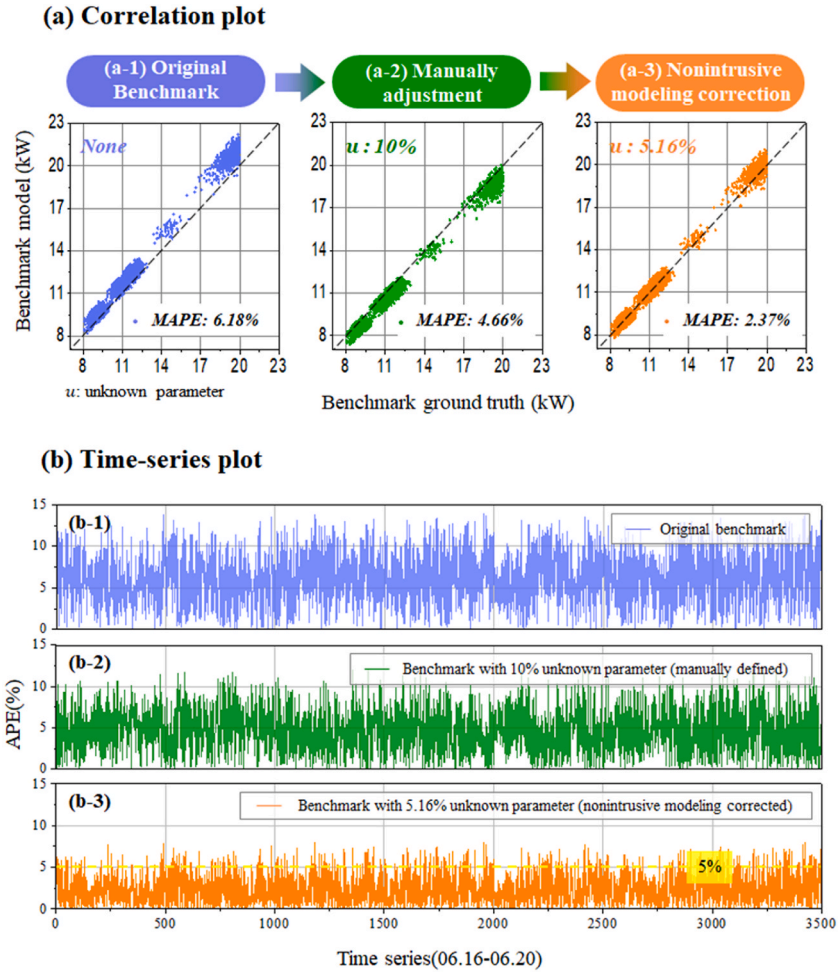


Fig. 7. Comparisons of the physics-based benchmark model accuracy with different unknown parameters.

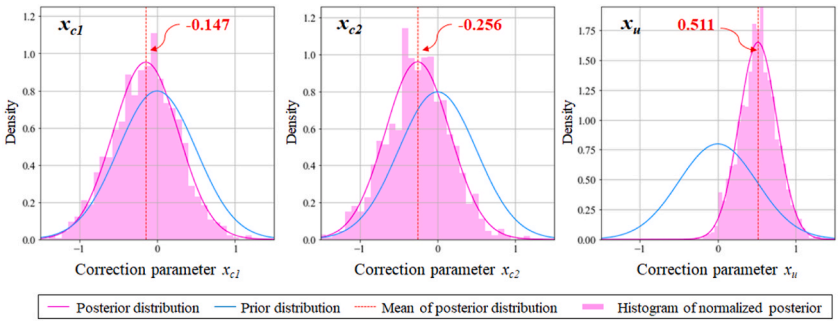


Fig. 8. Correction parameters of the physics-based model determined by Bayesian MCMC.

benchmark model. These results demonstrate the superior capability of the physics-based nonintrusive model in accurately adjusting the unknown parameter while correcting the target model and highlight the potential for advanced applications involving simultaneous corrections throughout system operations.

To analyze the usefulness of u, physics-based nonintrusive correction models were constructed using the three benchmark models in Fig. 7, and the accuracy of the corrected models was compared, as shown in Fig. 10. Without applying any unknown parameter, the poorest correction performance was observed, with the highest RMSE and MAPE values of 0.081 m^3/h and 3.37 %, respectively. Accuracy improvements of 30.3 % in MAPE values were achieved by incorporating the manually defined u. The best correction results were obtained using the benchmark model with nonintrusive modeling corrected u (Fig. 9), achieving a minimum RMSE of 0.034 m^3/h

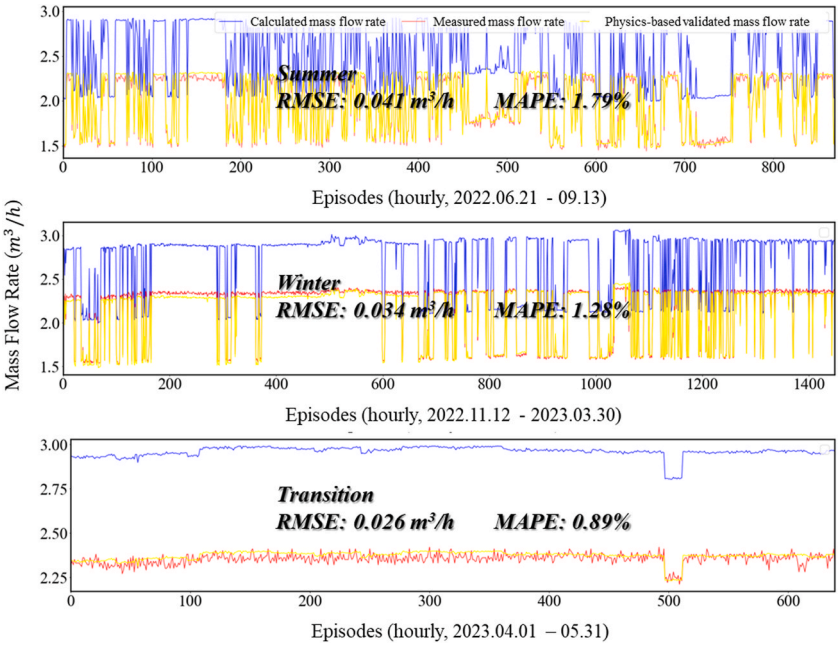


Fig. 9. Long-term validation results of the physics-based corrected target model.

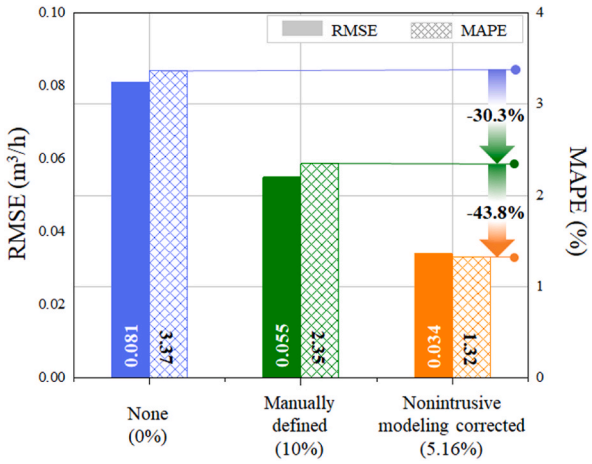


Fig. 10. Correction accuracy improvements by applying the unknown parameter.

and a significant overall improvement of 60.8 % in MAPE values. These results demonstrate the effectiveness of incorporating the unknown parameter when modeling with physics-based approach and highlight the importance of accurately correcting *u* through the modeling processes.

4.4. Monthly comparisons between the developed virtual models in a year

The monthly performance and accuracy of the developed models were compared and evaluated, as depicted in Figs. 11 and 12. Both data-driven and physics-based corrected models achieved significant reductions in RMSE and MAPE values. However, the data-driven corrected model displayed inconsistent performance, with uncertainty rates nearly doubling from November to May. As shown in Fig. 12, both models exhibited a strong linear relationship with the ground truth values from June to September, closely aligning with the diagonal. This alignment indicates high accuracy for both corrected models during this period. In contrast, during November to March, a significant error trend was observed in the data-driven corrected outputs. These outputs systematically deviated from the ground truth, with underestimations at lower ground truth values and overestimations at higher ground truth values. Furthermore, underestimations by the data-driven model were evident in April and May, further highlighting the limitations of the data-driven approach for long-term dynamic modeling. Conversely, the physics-based corrected models maintained high correction accuracy

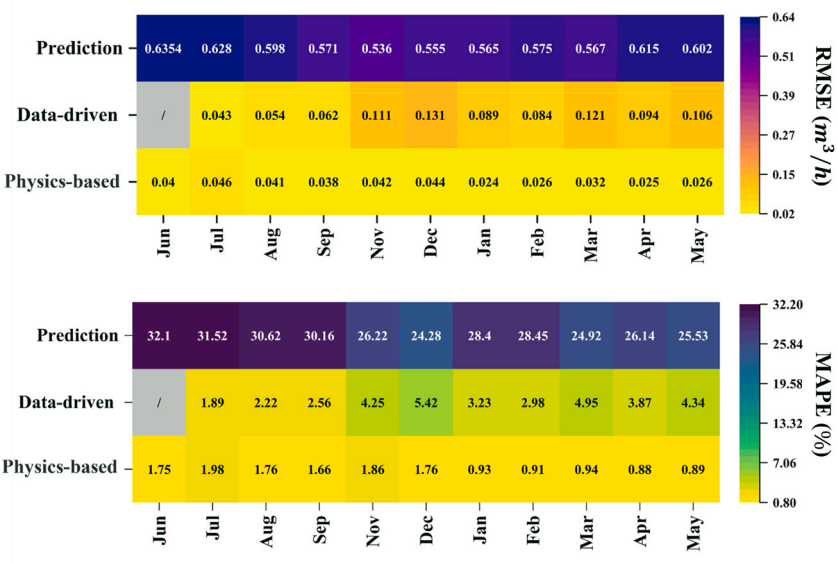


Fig. 11. Monthly RMSE and MAPE values of the developed virtual models.
Note: October was excluded due to extended public holidays and a lack of occupants.

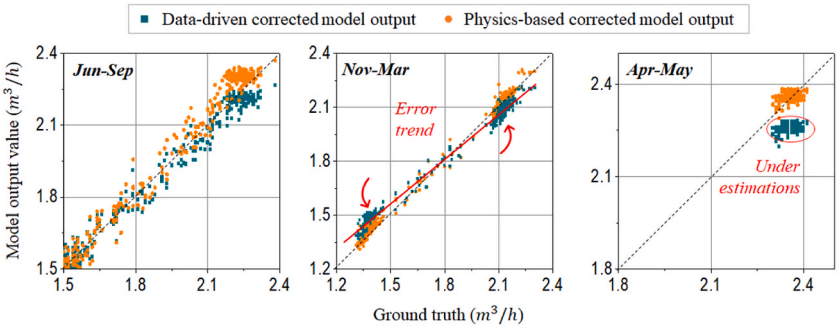


Fig. 12. Performance comparisons of data-driven and physics-based corrected models.

throughout the entire validation period, demonstrating their superior capability to provide reliable long-term estimates of the unknown target variable.

Additionally, the physics-based approach significantly improved modeling efficiency by eliminating the need for training and testing processes, reducing the overall modeling duration to less than a week and enabling an extended validation period.

The quality of the input datasets is critical to the development of physics-based models, as inaccurate or insufficient information may lead to poor model performance. The data-driven approach, with its capability to provide reliable outputs using limited input variables, can be applied to mitigate this issue and enhance the performance of physics-based models. This potential integration paves the way for the development of more advanced virtual models and supports intelligent applications in digital twin-enabled environments.

4.5. Implementation of nonintrusive modeling using Dynamo

The proposed virtual models were visualized in Revit as part of a digital twin environment, utilizing a workflow implemented through the Dynamo interface, as illustrated in Fig. 13. Within the Dynamo, the modeling framework, including design logic and interactions, was organized into five components: 1) Importing sensor information. According to the requirements of the virtual model construction, data and information from BAS system are imported in real-time into the Dynamo interface. 2) Loading the prediction model. Python scripts for various needs are used in the Dynamo interface to construct the prediction model. First, the real-time imported data is preprocessed to remove outliers and singularities, and then the processed data is imported into the model construction script to obtain the output of the prediction model. 3) Constructing benchmark models. Different Python scripts are used to construct physics-driven and data-driven benchmark models as needed for this study. During the benchmark model construction process, outputs from the prediction model are used as inputs to achieve nonintrusive calibration. Additionally, preprocessing of other input

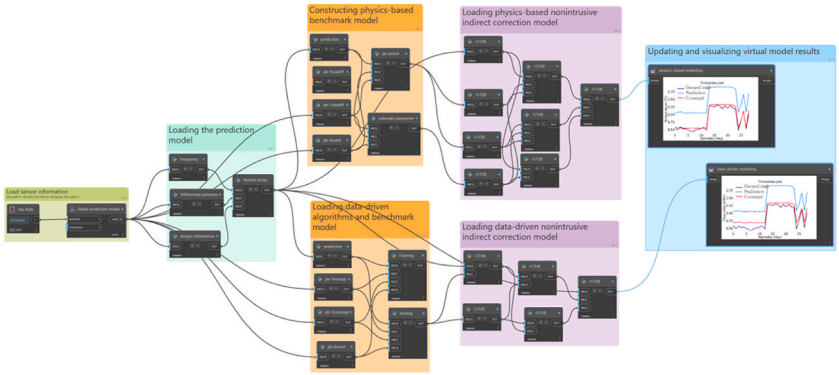


Fig. 13. Visualized modeling description through Dynamo interface.

variables based on Python scripts is performed during this construction phase. 4) Processing correction models. In the Dynamo interface, outputs from the prediction and benchmark models are fed into the Python script to construct the required correction model. Otherwise, the correction model script includes the Bayesian MCMC algorithm, which allows for real-time compensation of the prediction model, thereby ensuring its accuracy. 5) Updating the virtual model outputs. The outputs of the correction model are visualized using tools within the Dynamo interface to reflect calibration results in real-time. Moreover, the model results are updated in the digital twin environment to ensure stable operation of the virtual model. This workflow demonstrates the integration of various modeling components into a cohesive framework, enabling real-time correction and visualization of virtual model outputs to support stable and reliable operations.

The integration of Revit with the Dynamo interface plays a crucial role in addressing the challenges associated with achieving real-time updates between physical models and virtual models. This combination enables seamless data synchronization and facilitates dynamic interactions between the two environments, ensuring that changes in the physical model are accurately reflected in the virtual counterpart. By overcoming the limitations of traditional modeling approaches, this implementation enhances the reliability and efficiency of digital twin environments, providing a robust framework for monitoring, analysis, and decision-making in real-time applications.

An example of implementing a BIM-based, digital twin-enabled environment was shown in Fig. 14, where the properties of the target model were automatically updated to the Revit model through metadata management. Python scripts for the mass flow rate prediction model, benchmark model, and correction model are pre-written and finalized, ensuring that these scripts do not require modification unless the operating conditions change. During the execution of the Dynamo interface, real-time data from the BAS is recorded and loaded into the pre-written models. The accuracy of the correction results is then observed using visualization tools integrated into the interface. In this process, as long as BAS system data is transmitted without errors or missing values, the entire system operates autonomously without the need for human intervention, enabling seamless and automatic operation.

In addition, the implementation relies on Python scripts executed within the Dynamo interface. A key advantage of the proposed method is its flexibility and adaptability; even when applied to other buildings, as long as the input and output parameters remain unchanged, no additional modifications to the code are required. For buildings with increased complexity, adjustments are limited to modifying the relevant algorithms to meet specific requirements. The overall architecture, including the construction of the prediction model, development of the benchmark model, processing of the correction model, and updating of the virtual model outputs, remains

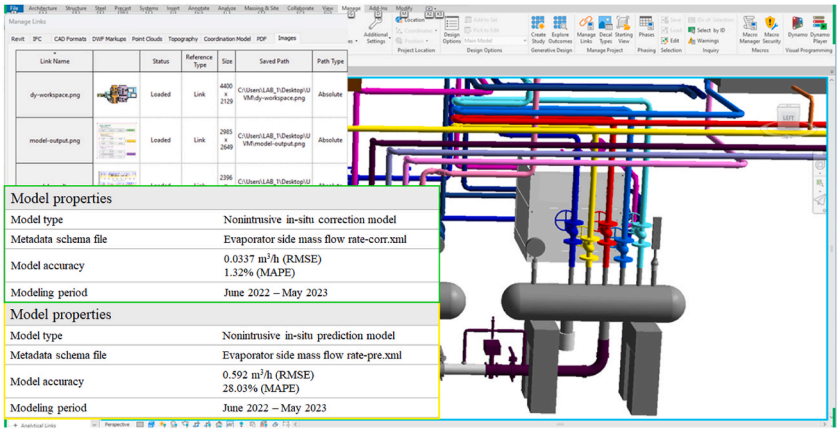


Fig. 14. Example of virtual model management with metadata.

intact. This modular design ensures scalability, efficiency, and ease of adaptation for diverse building scenarios without necessitating extensive reconfiguration of the system.

To further enhance the integration of BIM and physical models, fostering interdisciplinary collaboration between BIM experts, HVAC engineers, and data scientists is essential. This can be achieved through several key steps. First, defining common goals and establishing effective communication channels will align objectives and ensure a shared understanding of project requirements. Second, standardizing data integration processes and workflows will facilitate seamless interoperability between disciplines and systems. Third, promoting cross-disciplinary training and workshops will build mutual expertise and foster a culture of collaboration. Finally, leveraging digital twins and advanced data analytics will enable real-time monitoring, simulation, and optimization, further bridging the gap between virtual and physical systems. These steps will collectively improve the efficiency and effectiveness of BIM and physical model integration in complex building systems.

5. Conclusion

Long-term virtual models have great importance for providing insightful data on unknown variables in dynamic, sensor-limited environments. This study presented a nonintrusive in-situ modeling method with broad applicability to real HVAC systems. The modeling process consists of three key steps of prediction, benchmarking and correction. In the absence of target physical observations, the relational variable R is introduced to formulate indirect distance functions and enable nonintrusive corrections. Using the proposed physics-based approach, benchmark model performance is enhanced by incorporating the unknown parameter u. The annual RMSE of the prediction model was reduced from 0.592 m^3/h to 0.034 m^3/h, and the MAPE decreased from 28.03 % to 1.32 % through a single correction process. Long-term modeling efficiency was significantly improved while minimizing costs. The strong modeling capability for correcting u was validated through analyses, and the modeling processes were visualized in the digital twin-enabled environment using Dynamo.

Thus, for the target building or HVAC systems, the developed physics-based virtual models can be applied as virtual sensors to provide unobserved data, enabling holistic monitoring and advanced applications for optimal operational control. Meanwhile, the data-driven method, with its capability to model using limited inputs, can serve as a supportive approach to estimate missing or faulty input values, thereby improving the performance of the physics-based virtual models.

In this research, a long-term reliable physics-based nonintrusive in-situ virtual model was developed through a three-step manual process. For future research, it is recommended to investigate advanced methodologies aimed at automating the modeling procedures. This can be achieved through the integration of cloud platforms and advanced data communication technologies, which would significantly enhance the efficiency, scalability, and applicability of the proposed virtual modeling methods. Such advancements would enable their deployment in more sophisticated intelligent building applications, extending their use across a wider range of systems and promoting the development of next-generation smart environments.

CRediT authorship contribution statement

Yuxin Li: Writing – original draft, Methodology, Investigation. **Jeyoon Lee:** Software, Methodology. **Jiteng Li:** Resources, Data curation. **Peng Wang:** Software, Resources, Data curation. **Sungmin Yoon:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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